Seat No: _____________ Enrollment No: ___________________

PARUL UNIVERSITY

FACULTY OF ENGINEERING & TECHNOLOGY

B.Tech. Summer 2021-22 Examination

Semester: 5 **Date: 19-04-2022**
Subject Code: 03105303 **Time: 10:30am to 01:00pm**
Subject Name: Theory of Computation **Total Marks: 60**

Instructions:
1. All questions are compulsory.
2. Figures to the right indicate full marks.
3. Make suitable assumptions wherever necessary.
4. Start new question on new page.

Q.1 **Objective Type Questions -** Part 1 to 5 is MCQ, Part 6 to 10 is Fill in the blanks and Part 11 to 15 is one word answer (All are compulsory) (Each of one mark) **(15)**

1. The language described by the regular expression (0+1)*0(0+1)*0(0+1)* over the alphabet {0 1} is the set of
   (a) All strings containing at least two 1's
   (b) All strings containing at least two 0's
   (c) All strings that begin and end with either 0's or 1's
   (d) All strings containing the substring 00

2. A Context sensitive grammar is
   A) Type-0 B) Type-1 C) Type-2 D) Type-3

3. S → aSa | bSb | a | b
   The language generated by the above grammar over the alphabet is the set of
   (a) All palindromes
   (b) All odd length palindromes
   (c) Strings that begin and end with the same symbol
   (d) All even length palindromes

4. Which of the following does not have left recursions?
   (a) Chomsky Normal Form
   (b) Greibach Normal Form
   (c) Backus Naur Form
   (d) All of the mentioned

5. A language is regular if and only if
   (a) accepted by DFA
   (b) accepted by PDA
   (c) accepted by LBA
   (d) accepted by Turing machine

6. The language accepted by a Push down Automata is ______________ language.

7. Transition function of NDFA maps δ = _________________

8. Minimum numbers of states required to construct DFA which accepts all the strings of a's and b's where every string starts with 'aba' is ________________.

9. Grammar that produce more than one derivation tree for same sentence is ___________

10. According to Arden's theorem if every regular expression is in the form of R= Q + RP then there is a unique solution is ______________

11. Every finite subset of non-regular set is regular. True or False?

12. Any regular language has an equivalent CFG. True or False?

13. Define Left Factoring.

14. If a language and its complement are both regular, the language is recursive. True or False?

15. All languages can be generated by context- free grammar. True or False?

Q.2 Answer the following questions. (Attempt any three) **(15)**

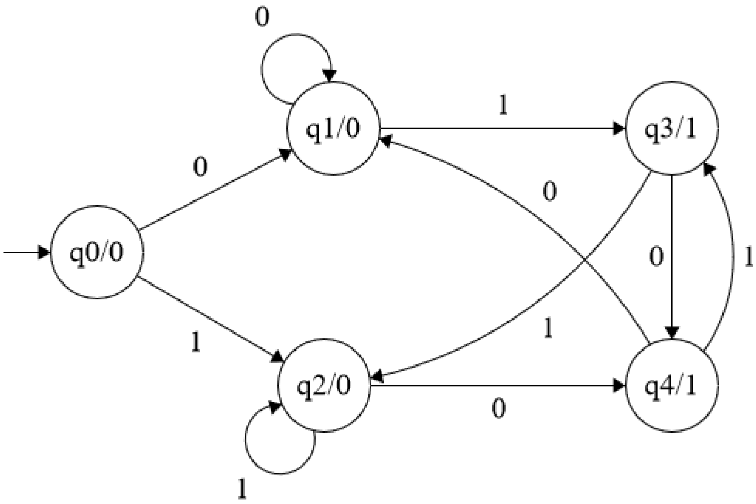
A) Using the principle of mathematical induction, prove that

$$1^2 + 2^2 + 3^2 + \ldots.. + n^2 = (1/6)\{n(n+1)(2n+1)\} \text{ for all } n \in N.$$

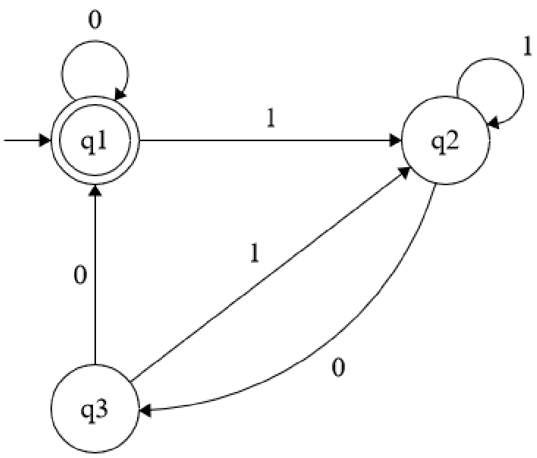
B) Convert the following CFG to CNF :

$S \rightarrow XYX$
$X \rightarrow aX \,/\, \varepsilon$
$Y \rightarrow bY \,/\, \varepsilon$

C) Convert to following Moore machine into Mealy machine.



D) Define ARDEN's Theorem. Construct a regular expression corresponding to the automata given below using ARDEN's Theorem.



Q.3 A) 1. Explain the types of Turing machine. **(3 Marks)** **(07)**

2. Using pumping lemma for regular sets prove that the language. **(4 Marks)**

$$L = \{0^n 1^m 0^{m+n} \mid m \geq 1 \text{ and } n \geq 1\} \text{ is not regular.}$$

B) 1. Explain Chomsky classification of grammar in detail. **(6 Marks)** **(08)**

2. Define Ambiguity in grammar. **(2 Marks)**

OR

B) Define Turing machine and Design a Turing Machine for the following Language: **(08)**

$L = \{a^n b^n \mid n \geq 0\}$

Q.4 A) Convert the following CFG into GNF **(07)**

$S \rightarrow aA \,/\, bB$

$A \rightarrow aA \,/\, \varepsilon$

$B \rightarrow bB \,/\, \varepsilon$

OR

A) Construct a PDA for the following Language: **(07)**

$L = \{a^i\, b^j\, c^k \mid i, j, k \geq 0 \text{ and } i + j = k\}$

B) Draw Deterministic Finite Automata for the following languages **(08)**

i) L1 = { x Є (0,1)* | x contains number of 1’s is multiple of 3}

ii) L2 = { x Є (0,1)* | x contains all the string in (0,1)* ending in 10 or 11}

iii) L3 = { x Є (0,1)* | x contains number 1’s is even and number of 0’s is even}

iv) L4 = { x Є (0,1)* | x contains at least one 1’s}